

Comprehensive Capstone Project Report: Enterprise Mutual Fund Analytics Platform

Course Tracking: Bluestock Data Science Internship

Project Classification: Final Capstone Engineering & Analytics Report

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Project Duration: 7-Day Intensive Capstone Cycle

Tools Used: Python, SQLite, Jupyter Notebook, Power BI, GitHub

Executive Summary

The Mutual Fund Analytics Platform was developed to analyze mutual fund performance, investor behavior, industry trends, and portfolio risk using data-driven techniques. The project integrates data engineering, analytics, visualization, and reporting into a single end-to-end workflow.

The platform processes multiple mutual fund datasets, performs extensive data cleaning and validation, computes performance and risk metrics, generates business insights, and presents results through an interactive Power BI dashboard.

The final solution supports decision-making for investors, analysts, and fund managers by providing visibility into fund performance, risk-adjusted returns, investor demographics, SIP trends, and portfolio concentration.

Problem Statement

Mutual fund investors often face challenges when evaluating investment options due to the large volume of available schemes and performance metrics.

Understanding risk, return, investor participation, and market trends requires significant analysis.

The objective of this project was to build a comprehensive analytics platform capable of:

- Processing mutual fund datasets.
- Measuring fund performance.
- Evaluating risk-adjusted returns.
- Studying investor behavior.
- Visualizing industry trends.
- Providing actionable investment insights.

Data Sources

The project utilized ten datasets representing different aspects of the mutual fund ecosystem.

Datasets Used

1. Fund Master Data
2. NAV History
3. AUM by Fund House
4. Monthly SIP Inflows
5. Category Inflows
6. Industry Folio Count
7. Scheme Performance
8. Investor Transactions
9. Portfolio Holdings
10. Benchmark Indices

The datasets collectively provide information regarding fund performance, investor activity, asset allocation, market trends, and industry participation.

Dataset	Purpose
Fund Master	Stores scheme metadata and AMFI details
NAV History	Historical NAV tracking
AUM Data	Assets Under Management analysis
SIP Inflows	Monthly SIP trend analysis
Category Inflows	Category-wise investment trends
Folio Count	Investor participation analysis
Scheme Performance	Returns and risk metrics
Investor Transactions	Investor behavior analysis
Portfolio Holdings	Portfolio diversification analysis
Benchmark Indices	Market comparison

Project Architecture

The project follows a structured analytics pipeline:

Raw Data → Validation → Cleaning → SQLite Database → Exploratory Analysis → Performance Analytics → Advanced Analytics → Dashboard Development → Final Reporting

The architecture ensures data quality, reproducibility, and scalability for future enhancements.

Data is gathered through a hybrid ingestion architecture. Static historical baselines are loaded from flat CSV files, while current valuations are fetched live via HTTP requests to the public MFAPI gateway.

Code:

```
import pandas as pd
```

```
import requests
```

```
import sqlite3
```

```
import logging
```

```
logging.basicConfig(level=logging.INFO, format='%(asctime)s -  
%(levelname)s - %(message)s')
```

```
def fetch_live_nav(scheme_code):
```

```
    """
```

```
    Queries the live MFAPI ecosystem to extract real-time financial data.
```

```
    Ensures that calculations reflect current market pricing.
```

```
    """
```

```
    url = f"https://api.mfapi.in/mf/{scheme_code}"
```

```
    try:
```

```
response = requests.get(url, timeout=10)
if response.status_code == 200:
    data = response.json()
    latest_nav = data['data'][0]['nav']
    latest_date = data['data'][0]['date']
    return float(latest_nav), latest_date
else:
    logging.warning(f'Failed connection for scheme: {scheme_code}. Code:
{response.status_code}')
    return None, None
except Exception as e:
    logging.error(f'Error querying live API for scheme {scheme_code}:
{str(e)}')
    return None, None
```

ETL Pipeline

Extraction

The first stage involved loading all provided CSV datasets and retrieving live mutual fund NAV information through MFAPI.

Key Activities

- CSV ingestion using Pandas.
- API integration for live NAV data.
- Validation of dataset completeness.

Transformation

Data cleaning and preprocessing steps included:

- Missing value handling.
- Duplicate removal.

- Date standardization.
- Transaction type normalization.
- NAV validation.
- Data quality verification.

Loading

Cleaned datasets were loaded into SQLite for structured querying and analytics.

Benefits of SQLite integration:

- Faster analytical queries.
 - Centralized storage.
 - Simplified reporting workflows.
-

Database Design

A relational database structure was created to support efficient analytics.

Primary tables included:

- Fund Master
- NAV History
- Investor Transactions
- Scheme Performance
- Portfolio Holdings

The database supports aggregation, trend analysis, and performance evaluation through SQL queries.

Exploratory Data Analysis (EDA)

Exploratory analysis was conducted to understand industry trends and identify patterns within the data.

NAV Trend Analysis

Historical NAV values showed long-term growth across major mutual fund schemes.

Market corrections were visible during periods of increased volatility; however, long-term appreciation remained evident.

AUM Growth Analysis

Assets Under Management increased steadily between 2022 and 2025, indicating growing investor participation and confidence in mutual fund investments.

SIP Trend Analysis

Monthly SIP inflows demonstrated continuous growth throughout the analysis period, highlighting increasing retail investor participation.

Category Inflow Analysis

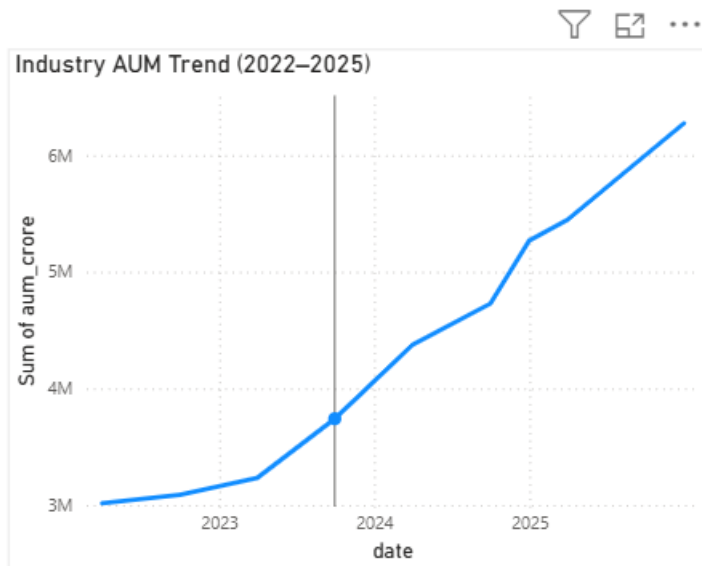
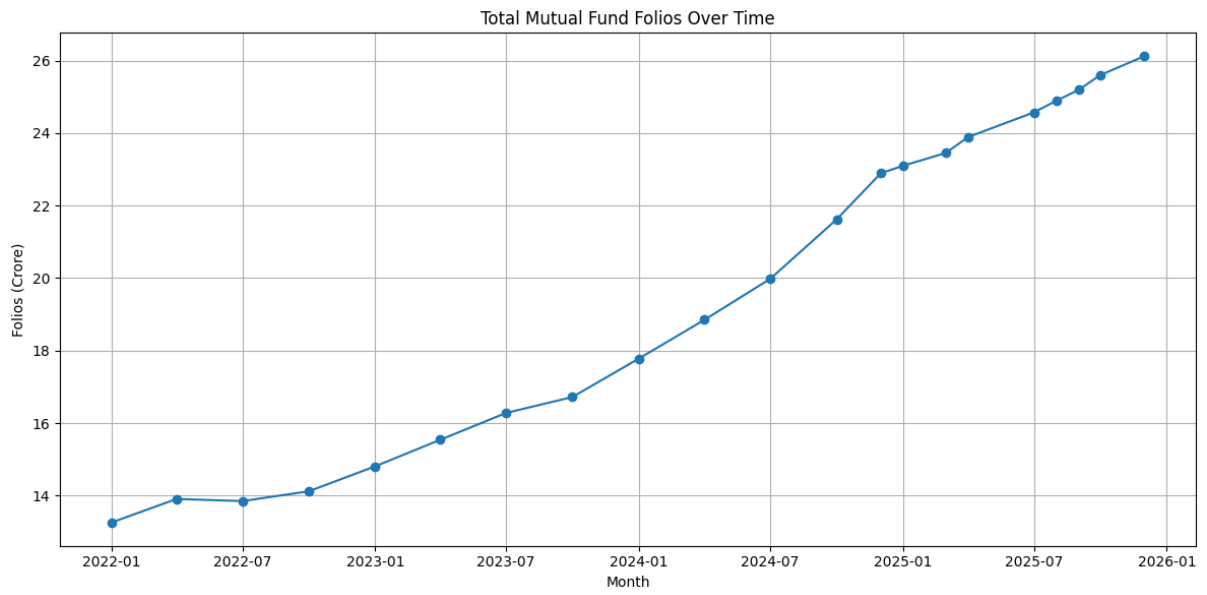
Certain fund categories consistently attracted higher inflows, indicating stronger investor preference toward diversified and growth-oriented strategies.

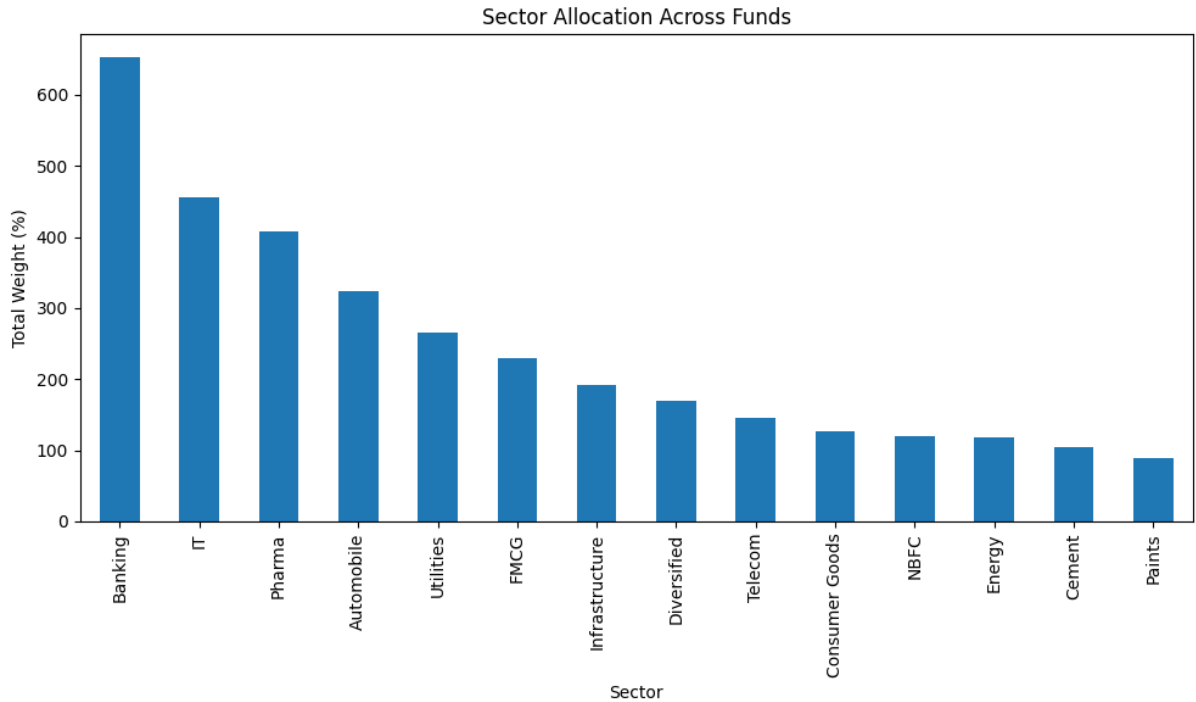
Investor Demographics

Investor activity was analyzed across:

- Age groups
- Gender
- States
- City tiers

This analysis revealed meaningful differences in participation patterns among demographic segments.





Performance Analytics

Performance evaluation focused on measuring return generation and risk-adjusted performance.

CAGR Analysis

Compound Annual Growth Rate was calculated to compare long-term fund performance.

Funds with stronger long-term growth consistently outperformed peers over multiple investment horizons.

$$CAGR = \left(\frac{V_{final}}{V_{initial}} \right)^{\frac{1}{t}} - 1$$

Where V_{final} is the final valuation, $V_{initial}$ is the initial capital baseline, and t is the elapsed time in fractional years.

Sharpe Ratio

The Sharpe Ratio measured risk-adjusted returns.

Funds with higher Sharpe Ratios demonstrated superior reward per unit of risk.

$$Sharpe = \frac{R_p - R_f}{\sigma_p}$$

Where R_p is the expected portfolio return, R_f is the risk-free rate of return (e.g., government treasury yield), and σ_p is the standard deviation of the portfolio's excess returns.

Sortino Ratio

Sortino Ratio was used to evaluate downside-risk-adjusted performance.

This metric provided additional insight into how funds behaved during adverse market conditions.

Alpha and Beta

Alpha measured excess return relative to the benchmark.

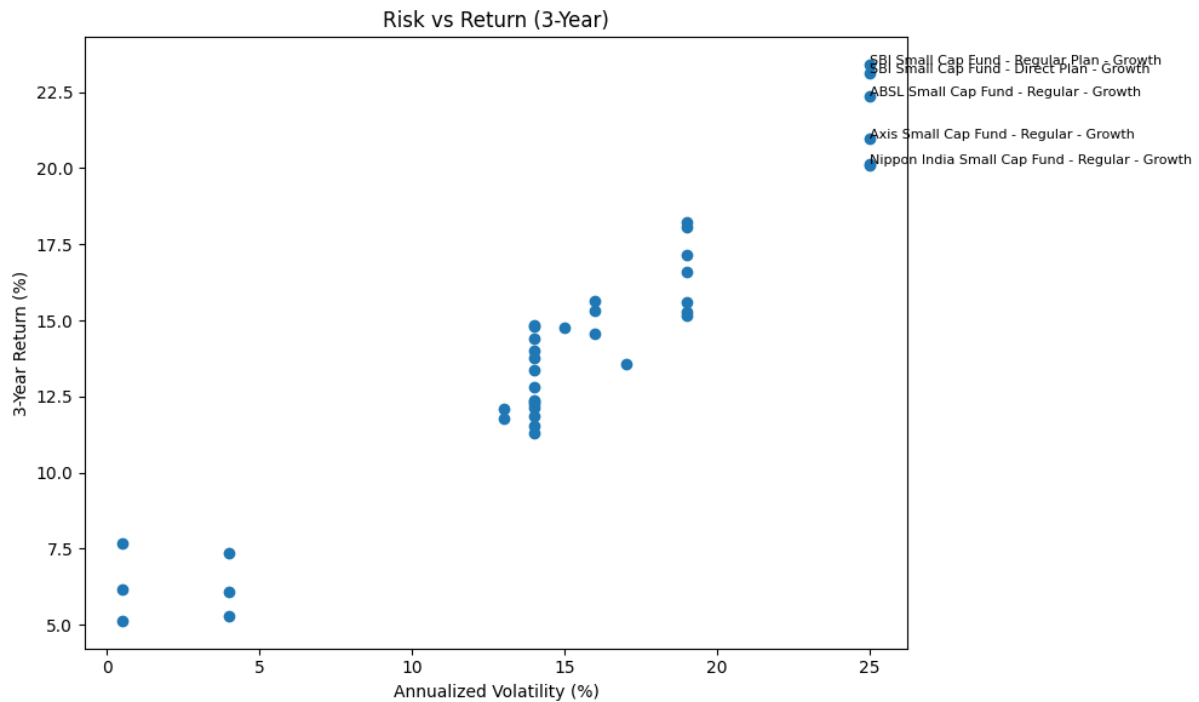
Beta quantified sensitivity to market movements.

These metrics helped identify funds capable of generating value beyond market performance.

Maximum Drawdown

Maximum Drawdown analysis identified the largest peak-to-trough declines experienced by funds.

This metric provided valuable insight into downside risk exposure.



Advanced Analytics

Advanced analytical techniques were implemented to deepen risk and investor behavior analysis.

Historical VaR and CVaR

Value at Risk (VaR) and Conditional Value at Risk (CVaR) were calculated for all schemes.

These metrics quantified potential losses under adverse market conditions and highlighted differences in downside risk profiles across funds.

Rolling Sharpe Ratio

A rolling 90-day Sharpe Ratio was computed to evaluate changing risk-adjusted performance over time.

The analysis revealed periods of stronger and weaker fund efficiency throughout market cycles.

Investor Cohort Analysis

Investors were grouped according to the year of their first transaction.

The analysis identified variations in:

- Average investment size.
- Total invested capital.
- Fund preferences.

SIP Continuity Analysis

Systematic Investment Plan continuity was assessed to identify investor retention patterns.

Investors with consistent SIP behavior represented a stable source of recurring inflows.

Fund Recommendation Engine

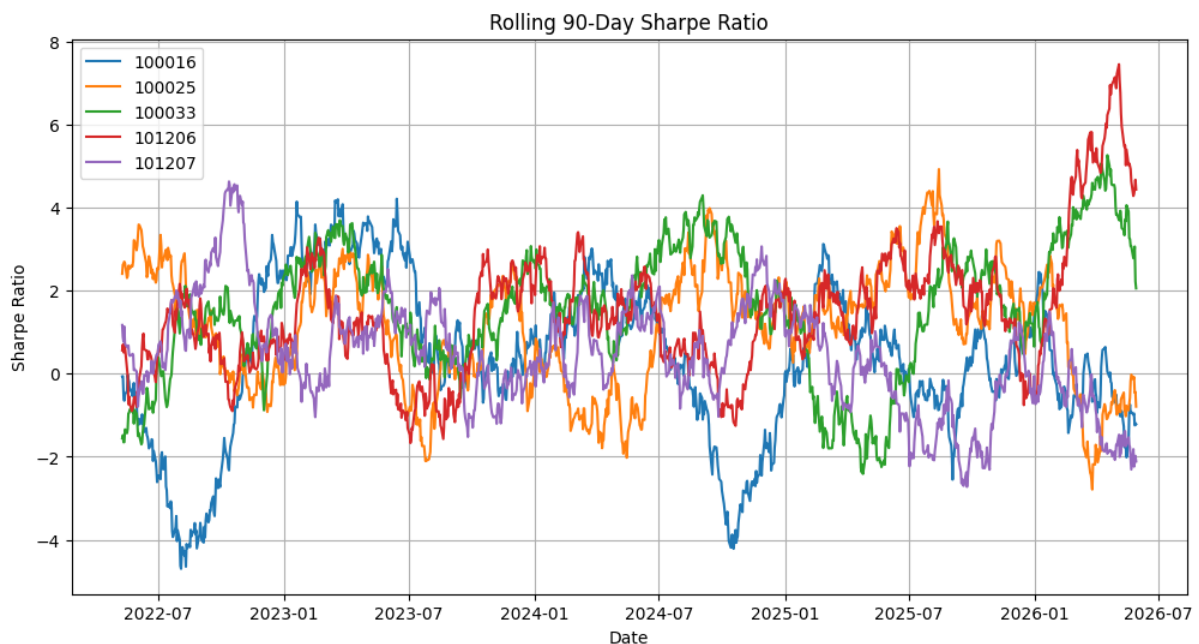
A rule-based recommendation system was developed using risk grades and Sharpe Ratios.

The system recommends suitable mutual fund schemes based on investor risk appetite.

Portfolio Concentration Analysis

The Herfindahl-Hirschman Index (HHI) was calculated to evaluate portfolio diversification.

Higher HHI values indicated concentrated portfolios, while lower values suggested broader diversification.



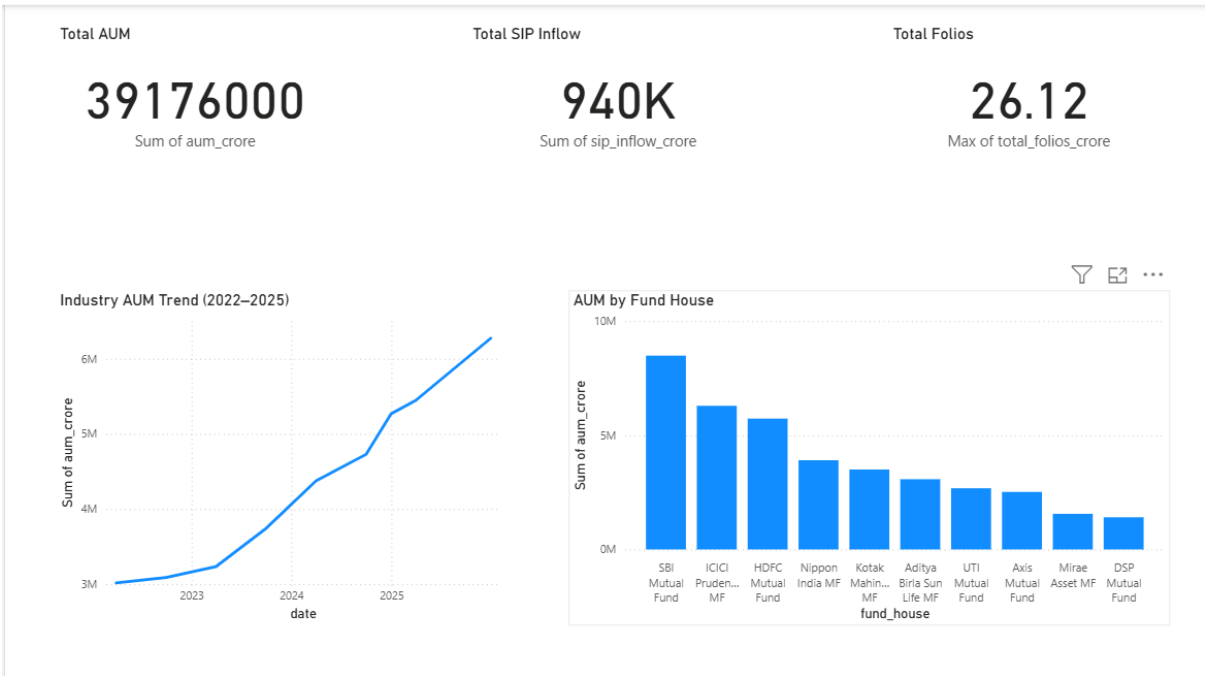
Dashboard Development

An interactive Power BI dashboard was developed to provide business-friendly visualizations.

Dashboard Pages

Industry Overview

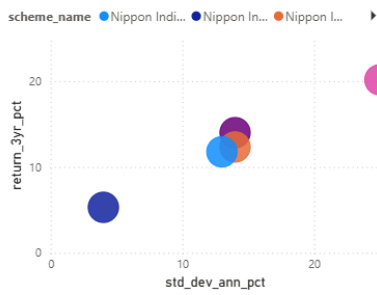
- Total AUM
- Total SIP Inflows
- Total Folios
- Industry Growth Trends



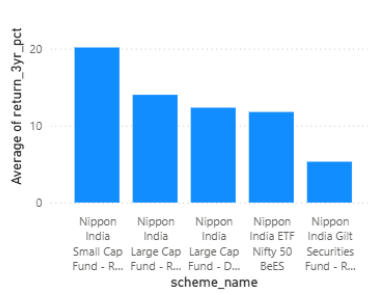
Fund Performance

- Risk vs Return Analysis
- Fund Scorecard
- Top Performing Funds

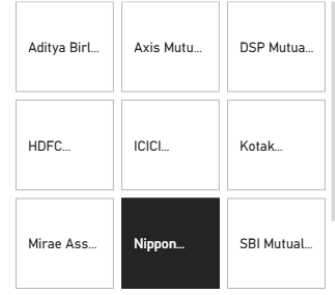
Risk vs Return (3-Year)



Top Funds by 3-Year Return



fund_house



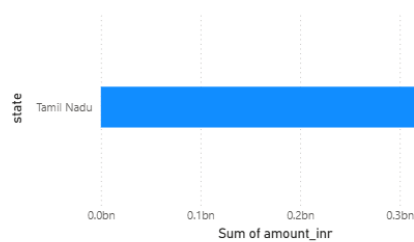
Fund Scorecard

scheme_name	fund_house	return_3yr_pct	sharpe_ratio	alpha	expense_ratio_pct	max_drawdown_pct
Nippon India Small Cap Fund - Regular - Growth	Nippon India MF	20.15	0.81	0.80	1.53	-30.87
Nippon India Large Cap Fund - Regular - Growth	Nippon India MF	14.00	1.00	0.86	1.51	-16.07
Nippon India Large Cap Fund - Direct - Growth	Nippon India MF	12.33	0.88	1.70	0.72	-18.14
Nippon India ETF Nifty 50 BeES	Nippon India MF	11.77	0.91	1.80	0.89	-26.75
Nippon India Gilt Securities Fund - Regular - Growth	Nippon India MF	5.31	1.33	0.89	0.55	-2.23

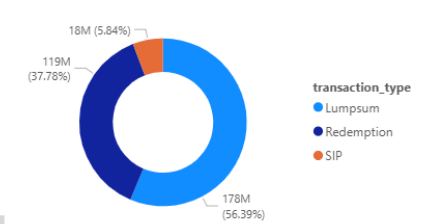
Investor Analytics

- State-wise Investments
- Gender Distribution
- Age Group Analysis
- Transaction Type Analysis

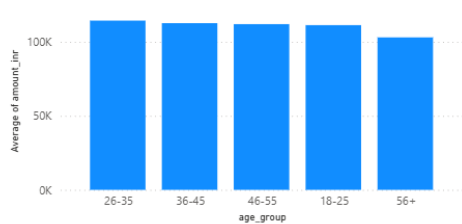
Transaction Amount by State



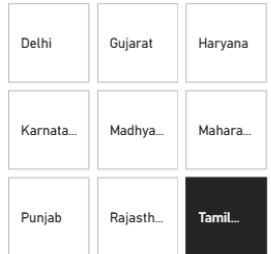
SIP vs Lumpsum vs Redemption



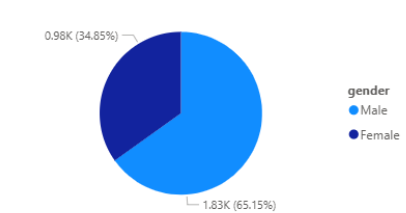
Average Investment by Age Group



state



Investor Gender Distribution



SIP & Market Trends

- SIP Growth Trends
- Category Inflow Trends
- Market Participation Metrics



Key Findings

1. Mutual fund participation increased steadily between 2022 and 2025.
2. SIP inflows demonstrated strong long-term growth, indicating increased retail investor confidence.
3. Risk-adjusted performance varied significantly across schemes, emphasizing the importance of Sharpe Ratio analysis.
4. Investor behavior differed across demographic groups and geographic regions.
5. Portfolio concentration levels varied substantially among funds, impacting diversification risk.

Recommendations

Based on the analysis:

- Investors should evaluate funds using both return and risk metrics.
 - SIP investing should be encouraged due to its consistency and long-term benefits.
 - Diversified portfolios should be preferred to reduce concentration risk.
 - Risk appetite should be matched with appropriate fund categories.
 - Continuous monitoring of performance metrics should be adopted for investment decisions.
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Limitations

The project has several limitations:

- Historical performance does not guarantee future returns.
 - Analysis depends on available datasets.
 - Recommendation engine uses rule-based logic rather than machine learning.
 - Market conditions may change over time.
 - Real-time dashboard refresh functionality is not implemented.
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Future Scope

Potential enhancements include:

- Real-time data ingestion.
- Machine learning recommendation systems.
- Cloud deployment.
- Automated dashboard refresh.
- Web-based analytics platform.
- Predictive performance forecasting models.

Conclusion

The Mutual Fund Analytics Platform successfully demonstrates an end-to-end data analytics workflow covering data engineering, exploratory analysis, performance evaluation, advanced risk metrics, dashboard development, and business reporting.

The project provides meaningful insights into mutual fund performance, investor behavior, and market trends while showcasing practical applications of Python, SQL, and Power BI in financial analytics.

The final solution serves as a scalable foundation for future investment analytics and decision-support systems.